IR devices that communicate with the controlled peripherals using infrared. A handful of remote controls also use radio signals. However, this has not proved popular, mainly because early radio TV remote controls would often switch channels on a neighbour's TV as well as on their owner's. As it does not pass through walls, most remote controls for electronic devices use an infrared diode. This diode emits a beam of light, typically with a wavelength of 940 nm. This beam of light reaches the device receptor and thereby controls it. This emitted beam of light is invisible to the human eye but not to video cameras, which see the diode as if it produces a purple light.

The presence of a carrier signal is used to trigger a function in the case of a single channel remote (single-function, one-button). There are many procedures that may be used in multi-function remotes. One of these consists in using signals of different frequency to modulate the carrier signal. The appropriate frequency filters are then applied to separate the signals after the demodulation of the received signal. However, these analogue methods are fast being replaced by their digital counterparts.

Thanks to the remote, you don't have to walk each time you want to switch channels on the television

An AM radio in very close proximity to a remote being operated is often an excellent means of hearing the signals being modulated on the infrared carrier.

In order to distinguish between the controls of devices from various companies, each company uses a different protocol to transmit infrared commands. Some of the different protocols that are employed are the different SIRCS versions used by Sony, the RC-6 from Philips, or the NEC TC101 protocol.

Remote controls are now extensively used in many areas including the military, space, industry, toys and video games.

There is no doubt that remote controls have transformed our lives and have greatly simplified the use and control of various household gadgets. However, there is a great increase in the number of gadgets in the average household and thus there is a related increase in the average number of remotes per household. A home may have up to eight different remotes at